

Are Emotions Arranged in a Circle? Geometric Analysis of Emotion Representations via Hyperspherical Contrastive Learning

情绪是环形排列的吗？通过超球面对比学习对情绪表示进行几何分析

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Abstract

摘要

Psychological research has long utilized circumplex models to structure emotions, placing similar emotions adjacently and opposing ones diagonally. Although frequently used to interpret deep learning representations, these models are rarely directly incorporated into the representation learning of language models, leaving their geometric validity unexplored. This paper proposes a method to induce circular emotion representations within language model embeddings via contrastive learning on a hypersphere. We show that while this circular alignment offers superior interpretability and robustness against dimensionality reduction, it underperforms compared to conventional designs in high-dimensional settings and fine-grained classification. Our findings elucidate the trade-offs involved in applying psychological circumplex models to deep learning architectures. Our code is available at <https://github.com/yama11235/EmpiricalCircumplexModel>

心理研究长期以来利用环形模型来结构情绪，将相似的情绪放置在相邻的位置，而对立的情绪则放置在对角线上。尽管这些模型经常被用来解释深度学习的表示，但它们很少被直接用于语言模型的表示学习，其几何有效性也未被探索。本文提出了一种方法，通过在超球体上进行对比学习，在语言模型的嵌入中诱导出环形的情绪表示。我们表明，尽管这种环形对齐在可解释性和对降维的鲁棒性方面具有优势，但在高维设置和细粒度分类任务中，其性能不如传统的设计。我们的研究结果阐明了将心理环形模型应用于深度学习架构时所涉及的权衡。我们的代码可在 <https://github.com/yama11235/EmpiricalCircumplexModel> 获取。

1 Introduction

1 引言

In recent years, the mechanistic interpretability of large language models (LLMs) has emerged as a pivotal field for ensuring AI safety and controllability (Zhao et al., 2024b; Bereska and Gavves, 2024; Opitz et al., 2025). Central to this field are the Linear Representation Hypothesis and Superposition Hypothesis, which posit that models represent human-interpretable concepts as linear directions within low-dimensional subspaces (Park et al., 2024; Elhage et al., 2022). This framework has been empirically validated through representation engineering, where intervening in specific directions allows for the direct manipulation of model behavior (Li et al., 2023; Arditì et al., 2024).

近年来，大型语言模型（LLMs）的机制可解释性已成为确保人工智能安全性和可控性的关键领域（Zhao 等，2024b；Bereska 和 Gavves，2024；Opitz 等，2025）。该领域的核心假设包括线性表示假设和叠加假设，这些假设认为模型将人类可解释的概念表示为低维子空间中的线性方向（Park 等，2024；Elhage

等, 2022)。这一框架通过表示工程得到了实证验证, 其中干预特定方向可以实现对模型行为的直接操控 (Li 等, 2023; Arditi 等, 2024)。

However, recent studies suggest that not all concepts are best captured by linear structures. Peri-
然而, 最近的研究表明, 并非所有概念都最好通过线性结构来表达。Peri-

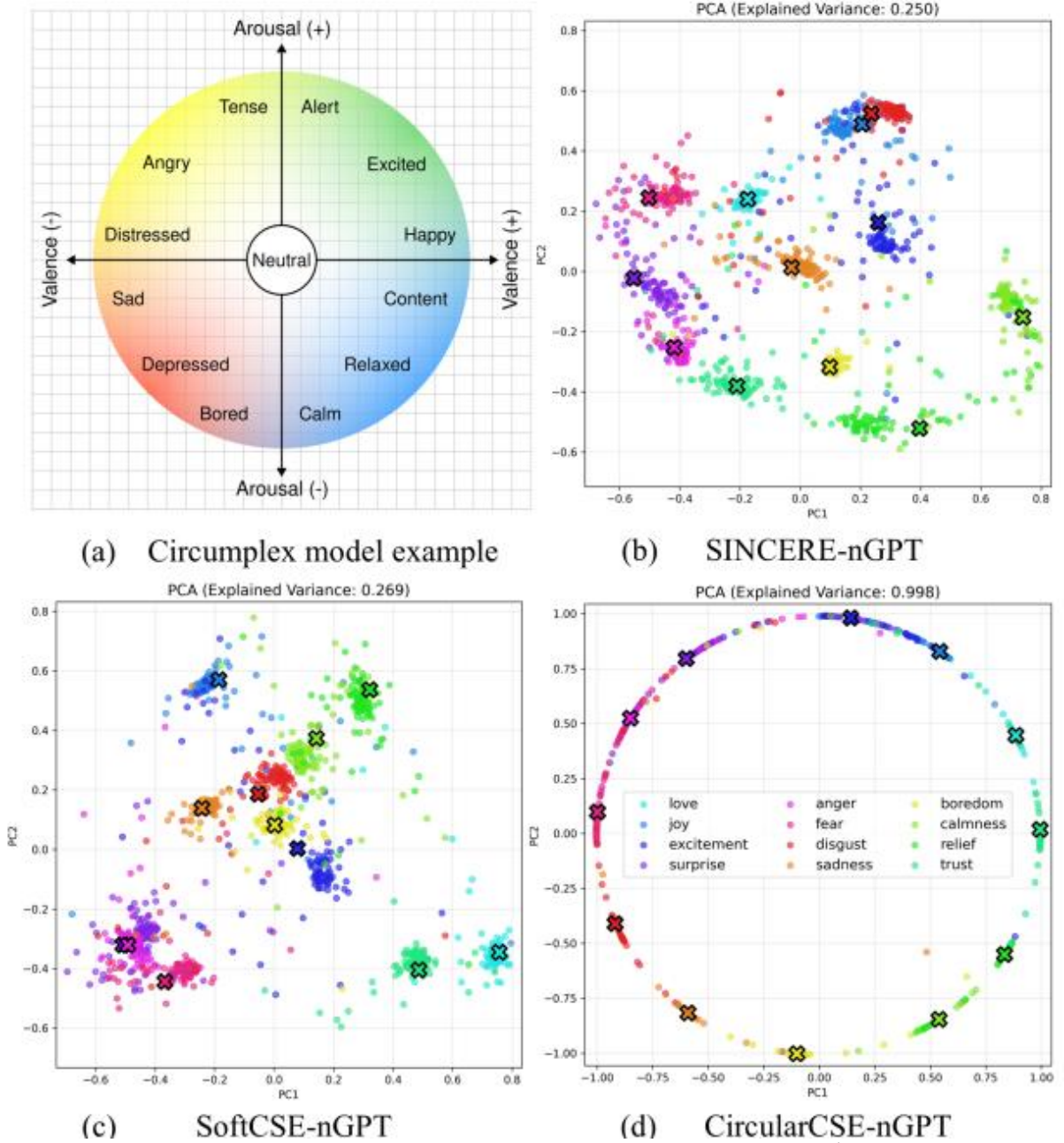


Figure 1: (a) An example of the psychological circumplex model of emotion¹. (b)(c)(d) PCA plots of the embeddings from the models trained in this study.

图 1: (a) 情绪的心理学区环模型示例¹。(b)(c)(d) 本研究中训练模型的嵌入值的主成分分析 (PCA) 图。

odic concepts, such as days of the week or months of the year, have been found to form circular embeddings, and models undergoing the “grokking” phenomenon in modular arithmetic ultimately arrange numerical representations in circular configurations (Engels et al., 2025; Park et al., 2025; Liu et al., 2022; Nanda et al., 2023). These findings raise a fundamental question: Are there other critical human concepts that are characterized by nonlinear, specifically circular,

manifold structures?

一些概念，如一周中的日子或一年中的月份，已被发现会形成环形嵌入，而经历“grokking”现象的模块化算术模型最终会将数值表示安排成环形结构（Engels 等，2025；Park 等，2025；Liu 等，2022；Nanda 等，2023）。这些发现引发了一个基本问题：是否还有其他关键的人类概念具有非线性、特别是环形的流形结构？

In psychology, the Circumplex Model of Affect has long proposed that emotions are arranged in a circle defined by two axes: valence and arousal (Figure 1(a)). Psychological emotion models are frequently employed in the analysis of machine learning models, with research investigating whether language models actually reflect these underlying structures (Wang and Zong, 2025; Zhao et al., 2025; Reichman et al., 2025). However,

在心理学中，情感的环形模型（Circumplex Model of Affect）长期以来提出，情绪是以两个轴（愉悦度和唤醒度）定义的一个圆圈来排列的（图 1(a)）。心理情感模型经常被用于分析机器学习模型，研究探讨语言模型是否实际上反映了这些潜在的结构（Wang 和 Zong, 2025; Zhao 等, 2025; Reichman 等, 2025）。然而，

since these analyses are often limited to post-hoc observations, such as centroid distances or label co-occurrence probabilities, they remain confined to analyzing general trends and do not extend to an analysis at the concept manifold level. Whether a language model expresses emotions in the same way humans do, or whether it should do so, cannot be rigorously discussed unless the manifold structure of the model’s embedding representations is explicitly reconstructed.

由于这些分析通常仅限于事后观察，例如质心距离或标签共现概率，因此它们仍然局限于分析一般趋势，而无法扩展到概念流形层面的分析。除非明确重建模型嵌入表示的流形结构，否则无法严谨地讨论语言模型是否以与人类相同的方式表达情感，或者是否应该这样做。

This paper investigates the validity of the circular emotion structure by explicitly inducing it within the embedding space. To this end, we first collect and synthesize a text dataset annotated with emotions assumed to follow a circular structure. We then train emotion representations using nGPT (Loshchilov et al., 2025), an architecture designed for hyperspherical representation learning, in combination with three contrastive loss functions: SINCERE (Feeney and Hughes, 2023), SoftCSE (Zhuang et al., 2024), and our proposed CircularCSE. These objectives calibrate distances between emotion labels according to psychologically grounded circular distances (CD) (Zhao et al., 2024a). Finally, we evaluate the resulting representations across three backbone categories (BERT-like models, LLM-based encoders, and decoder-only LLMs) using a comprehensive set of metrics, including V-Measure for discriminative power and CD-r for psychological alignment.

本文研究了循环情绪结构的有效性，通过在嵌入空间中显式地诱导该结构来实现。为此，我们首先收集并合成一个带有情绪标注的文本数据集，这些情绪被假设遵循循环结构。接着，我们使用 nGPT (Loshchilov 等，2025)，一种用于超球面表示学习的架构，结合三种对比损失函数：SINCERE (Feeney 和 Hughes, 2023)、SoftCSE (Zhuang 等，2024) 以及我们提出的 CircularCSE，来训练情绪表示。这些目标根据基于心理的循环距离 (CD) (Zhao 等，2024a) 校准情绪标签之间的距离。最后，我们使用一系列综合指标，包括用于判别能力的 V-Measure 和用于心理对齐的 CD-r，评估由此得到的表示在三个主干类别（BERT 类模型、基于 LLM 的编码器和仅解码器的 LLM）上的表现。

Our results reveal a stark structural dilemma. While the circular structure (CircularCSE) offers superior interpretability and remains robust in low-dimensional spaces or with few labels (Figure 1(d)), its performance degrades significantly in high-dimensional or fine-grained label settings compared to conventional designs. We provide a theoretical explanation for this: SINCERE thrives in high dimensions by arranging labels as an orthogonal simplex (90° margins), whereas CircularCSE’s 2D ring geometry forces much tighter boundary margins, inherently limiting discriminability as the number of labels increases.

我们的结果揭示了一个明显的结构性困境。虽然循环结构（CircularCSE）在可解释性方面具有优势，并且在低维空间或标签较少的情况下保持稳健（图 1(d)），但与传统设计相比，在高维或细粒度标签设置下其性能会显著下降。我们对此提供了理论解释：SINCERE 通过将标签排列为正交单纯形（90° 边距）在高

维空间中表现出色，而 CircularCSE 的二维环形几何结构则强制要求更紧密的边界边距，随着标签数量的增加，这种结构本质上限制了可区分性。

This conflict suggests that aligning models with human interpretation, which implicitly assumes low-dimensional manifold structures, comes at a structural cost to discriminative power. However, aligning model representations with human psychology offers diverse benefits, ranging from intuitive mechanistic interpretation to reduced computational costs. Our research re-examines the validity of incorporating human interpretability into model design from the perspective of deep learning.

这种冲突表明，将模型与人类解释对齐，这隐含地假设了低维流形结构，会以牺牲判别能力为代价。然而，将模型表示与人类心理对齐可以带来多种好处，从直观的机制解释到降低计算成本。我们的研究从深度学习的角度重新审视了在模型设计中纳入人类可解释性的有效性。

2 Related Work

2 相关工作

Psychological models of emotion are generally classified based on whether they represent emotion in a continuous or discrete space. A representative example of a continuous model is Russell's Circumplex Model of Affect, which posits that emotions are represented along two axes: valence and arousal (Russell, 1980). Along with the PAD model, which adds a dominance axis, it has been widely adopted in recent years (Mehrabian and Russell, 1974). In contrast, discrete emotion models construct a space from a set of basic emotions, where diverse emotional states are expressed through the compounding or subdivision of these basics. Plutchik proposed the "Wheel of Emotions," consisting of eight basic emotions (Plutchik, 1980). Regardless of the continuous or discrete space, most foundational psychological models represent emotions as a circular structure (Shaver et al., 1987; Ekman, 1992). This design reflects the clear similarities (e.g., joy and excitement) and polarities (e.g., positive and negative) inherent in emotions.

情绪的心理学模型通常根据其是否将情绪表示在连续空间或离散空间中进行分类。一个典型的连续模型例子是 Russell 的情绪环形模型 (Circumplex Model of Affect)，该模型认为情绪沿着两个轴进行表示：效价 (valence) 和唤醒度 (arousal) (Russell, 1980)。与添加了支配轴 (dominance axis) 的 PAD 模型一起，该模型近年来被广泛采用 (Mehrabian and Russell, 1974)。相比之下，离散情绪模型从一组基本情绪构建空间，通过这些基本情绪的组合或细分来表达多样化的情绪状态。Plutchik 提出了“情绪轮” (Wheel of Emotions)，包含八种基本情绪 (Plutchik, 1980)。无论是在连续空间还是离散空间，大多数基础心理学模型都将情绪表示为环形结构 (Shaver et al., 1987; Ekman, 1992)。这种设计反映了情绪中固有的清晰相似性 (例如喜悦和兴奋) 和对立性 (例如积极与消极)。

Geometry of LLM Representations. While the analysis of emotion representations in LLMs is common, many studies focus on examining the macroscopic relationships between emotions in existing models based on co-occurrence probabilities or the Euclidean distances between centroids (Guo and Choi, 2021; Wang and Zong, 2025; Zhao et al., 2025; Reichman et al., 2025). Outside the domain of emotion, it has been discovered that periodic concepts and numerical values in modular addition tasks exhibit circular structures (Engels et al., 2025; Park et al., 2025; Liu et al., 2022; Nanda et al., 2023). However, these findings are limited to specific conditions, such as those involving sparse autoencoders (Bricken et al., 2023) or grokking. To the best of our knowledge, no study has successfully constructed a circular structure for emotion representations within a standard language model.

LLM 表示的几何结构。虽然对 LLM 中情感表示的分析很常见，但许多研究侧重于基于共现概率或质心之间的欧几里得距离来考察现有模型中情感之间的宏观关系 (Guo 和 Choi, 2021; Wang 和 Zong, 2025; Zhao 等, 2025; Reichman 等, 2025)。在情感领域之外，人们发现模块加法任务中的周期性概念和数值值呈现出环形结构 (Engels 等, 2025; Park 等, 2025; Liu 等, 2022; Nanda 等, 2023)。然而，这些发现仅限于特定条件，例如涉及稀疏自编码器 (Bricken 等, 2023) 或 grokking 的情况。据我们所知，尚无研究成功在标准语言模型中构建出情感表示的环形结构。

Design of Contrastive Learning. Although many studies employ improved or custom variants of contrastive loss functions, they are primarily designed to enhance discriminative performance, with few focusing on the explicit induction of manifold structures (Choi et al., 2020; Yang et al., 2021; Deng et al., 2022). In the domain of image emotion classification, a prior study has incorporated angular

对比学习的设计。尽管许多研究采用改进或定制的对比损失函数变体，但它们主要设计用于增强判别性能，而很少关注显式诱导流形结构（Choi 等，2020；Yang 等，2021；Deng 等，2022）。在图像情感分类领域，先前的研究已经引入了角度

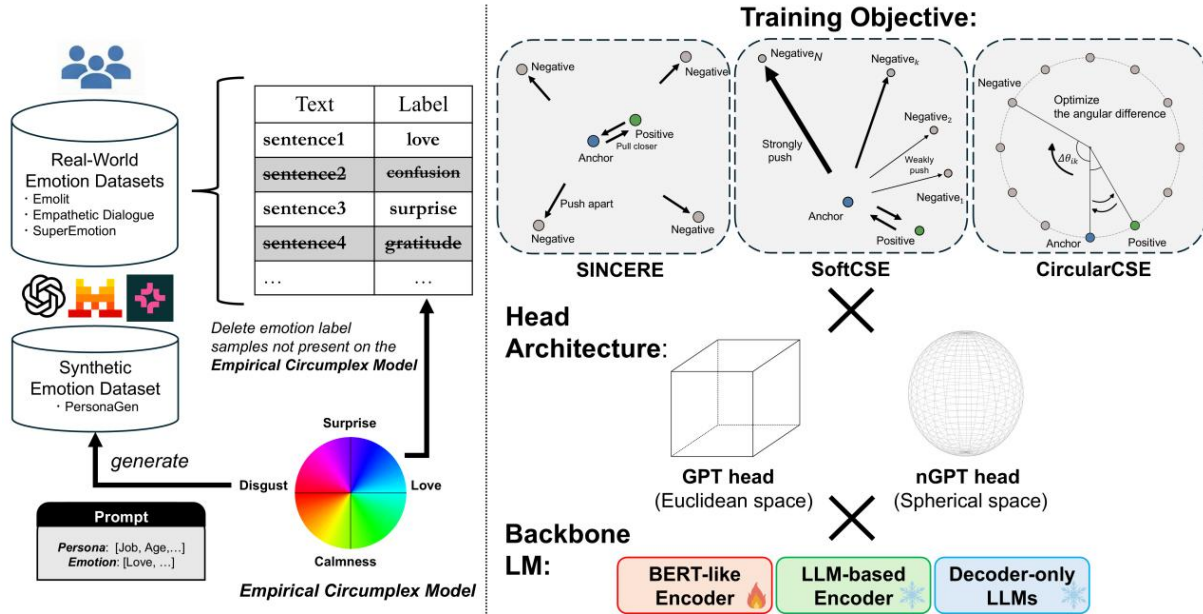


Figure 1: images/3848389cfcf23e7416b94b851e29ac6d4765f59edeff286020614d01a808f4ce.jpg

Figure 2: Overview of our experimental framework. (Left) Dataset construction procedure. Corresponding emotion labels are extracted or synthesized to reproduce the circumplex emotion structure. (Right) Training of GPT or nGPT heads across three backbone architectures using three distinct loss functions.

图 2：我们的实验框架概述。（左）数据集构建流程。相应的感情标签被提取或合成以重现环行情绪结构。（右）使用三种不同的损失函数，在三种不同的主干架构上训练 GPT 或 nGPT 头。

differences based on a circular assumption into the loss function (Yang et al., 2021). However, because this method operates within Euclidean space and combines this term with other loss functions, it is highly probable that the resulting manifold structure does not converge to a true circle.

基于循环假设的差异被引入损失函数中（Yang 等，2021）。然而，由于该方法在欧几里得空间内运行，并将该项与其他损失函数结合，所得到的流形结构很可能无法收敛到真实的圆。

In this study, we design a loss function that induces a circular structure on a hypersphere, thereby explicitly reproducing this geometry within the language model’s embedding space. This allows us to rigorously discuss the utility and validity of applying psychological circumplex models of emotion to deep learning.

在本研究中，我们设计了一个损失函数，使其在超球体上诱导出圆周结构，从而在语言模型的嵌入空间中明确地再现这种几何结构。这使我们能够严格地讨论将情绪的心理圆周模型应用于深度学习的效用和有效性。

3 Methodology

3 方法论

3.1 Overview

3.1 概述

Figure 2 presents an overview of our experimental framework. We collect datasets where emotions are assumed to be equally spaced on a circle and train the model to encode their representations. To replicate the circular structure, we employ the following two learning strategies:

图 2 展示了我们的实验框架概览。我们收集数据集，其中情绪被假定在圆周上等间距分布，并训练模型以编码其表示。为了复制圆周结构，我们采用以下两种学习策略：

Geometry of the embedding space. By learning emotion representations in a spherical space, we encourage the model to reproduce the circular structure, ensuring that differences are represented solely by angle distance (Section 3.3).

嵌入空间的几何结构。通过在球形空间中学习情感表示，我们鼓励模型重现圆形结构，确保差异仅通过角度距离来表示（第 3.3 节）。

Contrastive learning design. We align the relationships between emotion labels by calibrating pairwise distances or gradient weights between an-

对比学习设计。我们通过校准情感标签之间的成对距离或梯度权重，来对齐情感标签之间的关系或

chor and negative samples according to their positions on the Circumplex Model of Affect (Section 3.4).

根据情感环形模型（第 3.4 节）中它们在情感环形模型中的位置，将样本分为积极和消极样本。

3.2 Preliminary

3.2 初步准备

Figure 3 illustrates the basic circumplex emotion arrangement employed in this study, which consists of 12 emotion categories (hereafter referred to as the Empirical Circumplex Model, or ECM). While our model is primarily based on Russell’s Circumplex Model of Affect (Russell, 1980; Yik et al., 2011), certain emotions have been substituted to align with the labels available in real-world datasets. For instance, although terms like “sleepy” or “quiet” would ideally represent the state of deactivation at $3\pi/2$, we adopt “calmness” due to the absence of datasets annotated with these specific labels. Formally, we define each emotion label as $y \in \mathcal{E}$, where y denotes the class of emotion and $\mathcal{E}$ denotes the complete set of emotion labels of size $E := |\mathcal{E}|$. In our experiments, we utilize a set of N text-label pairs for a given emotion classification dataset, denoted as $D = \{(x_i, y_i) \mid y_i \in \mathcal{E}\}_{i=1}^N$.

图 3 展示了本研究中使用的基于环形情感排列模型，该模型包含 12 种情感类别（以下简称实证环形模型，或 ECM）。虽然我们的模型主要基于 Russell 的情感环形模型（Russell, 1980; Yik 等, 2011），但为了与现实世界数据集中的标签相匹配，某些情感已被替换。例如，尽管像“困倦”或“安静”这样的术语理想情况下应代表 $3\pi/2$ 的去激活状态，但由于缺乏标注这些特定标签的数据集，我们采用“平静”作为替代。正式而言，我们将每个情感标签定义为 $y \in \mathcal{E}$ ，其中 y 表示情感类别， $\mathcal{E}$ 表示情感标签的完整集合，其大小为 $E := |\mathcal{E}|$ 。在我们的实验中，我们使用一组 N 文本-标签对，用于给定的情感分类数据集，记作 $D = \{(x_i, y_i) \mid y_i \in \mathcal{E}\}_{i=1}^N$ 。

3.3 Head Architecture

3.3 头部架构

We append a single Transformer block (projection head) to a pre-trained backbone, utilizing the head's output embedding space to represent the geometric structure of emotions. Let t denote the index of the token in the input sequence $[1, \dots, T]$ and d denote the dimension of the model. A standard

我们在一个预训练的主干网络上添加一个单一的 Transformer 块（投影头），利用该头部的输出嵌入空间来表示情感的几何结构。设 t 表示输入序列 $[1, \dots, T]$ 中的标记索引， d 表示模型的维度。一个标准的

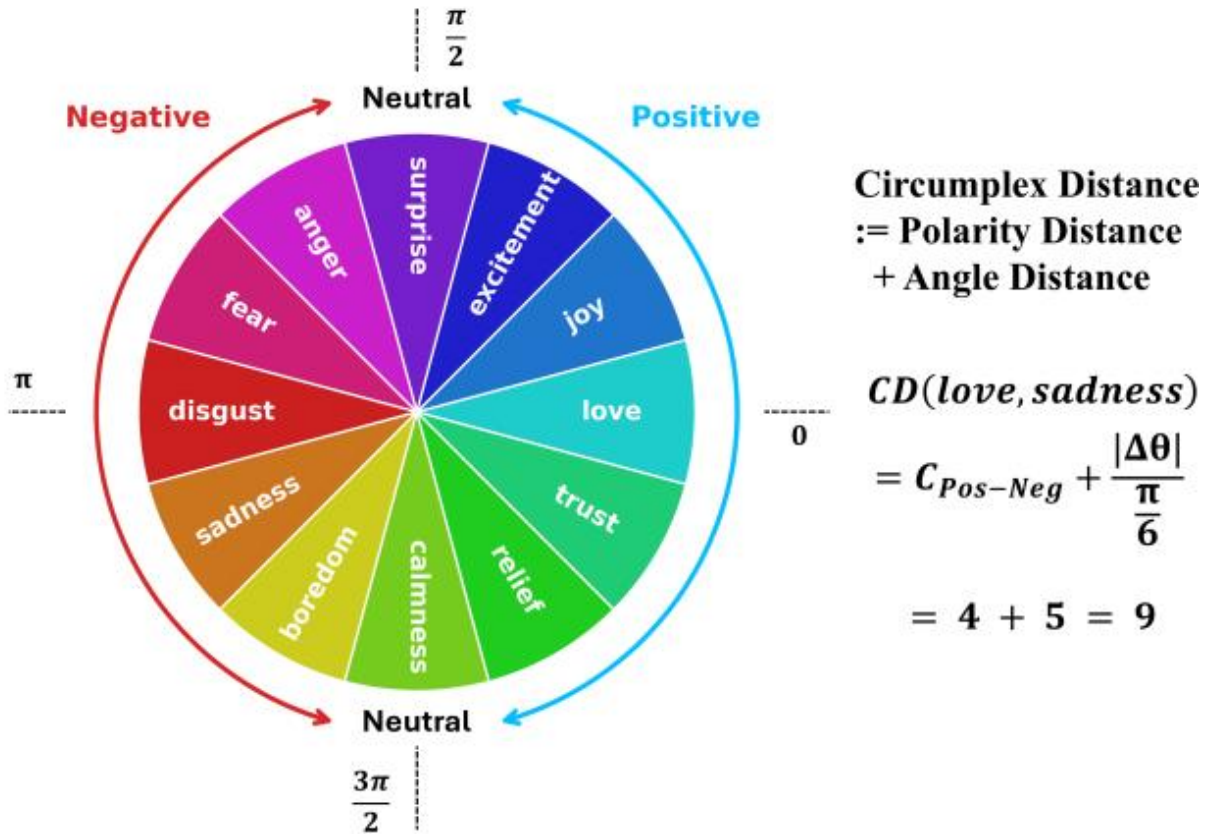


Figure 2: images/8e5ae5c2af682ceafc0f9f258bc0f9d9eec3a741d0bf5dbb3a014233e962968d.jpg

Figure 3: Our empirical circumplex model (ECM) and definition of Circumplex Distance (CD).

图 3：我们的经验环形模型（ECM）和环形距离（CD）的定义。

Transformer block consists of an attention mechanism (ATTN), a multi-layer perceptron (MLP), and normalization modules (RMSNorm), formulated as follows:

Transformer block 包含一个注意力机制（ATTN）、一个多层感知机（MLP）和归一化模块（RMSNorm），其公式如下：

$$h'_t = h''_t + \text{ATTN}(\text{RMSNorm}(h''_t)), \quad (1)$$

$$h_t = h'_t + \text{MLP}(\text{RMSNorm}(h'_t)),$$

where $h_t, h'_t, h''_t \in R^d$. Here, h''_t denotes the backbone output, which serves as the head input, and h'_t denotes the head output. This space corresponds to a d-dimensional Euclidean space where the arrangement of embeddings is unconstrained. Since the norm itself carries semantic meaning, it is inherently difficult to induce a circular geometry. Therefore, we adopt the normalized Transformer Block (nGPT), an architecture explicitly designed for spherical space (Loshchilov et al., 2025). The operations within the nGPT block are defined as follows:

这里 $h_t, h'_t, h''_t \in R^d$ 。在这里, h''_t 表示主干网络的输出, 它作为头部网络的输入, 而 h'_t 表示头部网络的输出。这个空间对应于一个 d 维欧几里得空间, 其中嵌入向量的排列不受限制。由于范数本身具有语义含义, 因此很难诱导出环形几何。因此, 我们采用归一化 Transformer 块 (nGPT), 这是一种专门设计用于球形空间的架构 (Loshchilov 等人, 2025 年)。nGPT 块中的操作定义如下:

$$h'_t = \text{Norm}((1 - \alpha_A) \odot \text{Norm}(h''_t) + \alpha_A \odot \text{Norm}(\text{ATTN}(h''_t))),$$

$$h_t = \text{Norm}((1 - \alpha_M) \odot \text{Norm}(h'_t) + \alpha_M \odot \text{Norm}(\text{MLP}(h'_t))), \quad (2)$$

where $\text{Norm}()$ represents ℓ_2 normalization, $h_t, h'_t \in S^{d-1}$, and $\alpha_A, \alpha_M \in R^d$ are learnable parameters. nGPT removes normalization modules from standard transformer blocks and instead normalizes all hidden states and weights to unit norm along the feature dimension. Consequently, since the output of each module resides on a hypersphere and is updated along geodesics, the optimization process within the block can be viewed as traversing the spherical manifold. The rationale for adopting the nGPT architecture and further details regarding the intra-block processing are provided in Appendices A and B. In our experiments, we derive the final sentence embedding e by applying a pooling operation to the sequence of hidden states $h_{1:T}$, followed by normalization:

其中 $\text{Norm}()$ 表示 ℓ_2 归一化, $h_t, h'_t \in S^{d-1}$ 和 $\alpha_A, \alpha_M \in R^d$ 是可学习的参数。nGPT 从标准的 transformer 块中移除了归一化模块, 而是沿着特征维度对所有隐藏状态和权重进行单位范数归一化。因此, 由于每个模块的输出位于超球面上, 并且沿测地线进行更新, 块内的优化过程可以视为在球面流形上进行遍历。采用 nGPT 架构的 rationale 以及块内处理的更多细节在附录 A 和 B 中提供。在我们的实验中, 我们通过对隐藏状态序列 $h_{1:T}$ 应用池化操作, 然后进行归一化, 得到最终的句子嵌入 e :

$$e = \text{Norm}(\text{Pooling}(h_{1:T})), \quad (3)$$

where $e \in S^{d-1}$, $h_{1:T} \in R^{T \times d}$. The specific pooling operation depends on the backbone model and is defined as follows:

where $e \in S^{d-1}$, $h_{1:T} \in R^{T \times d}$ 。具体的池化操作取决于主干网络模型, 定义如下:

$$\text{Pooling}_{\text{cls}} := h_1, \text{Pooling}_{\text{last}} := h_T, \text{Pooling}_{\text{mean}} := \frac{1}{T} \sum_{t=1}^T h_t. \quad (4)$$

For comparison, a conventional Transformer block (GPT head) is also trained as a baseline.

作为对比, 一个传统的 Transformer 块 (GPT 头) 也作为基线进行训练。

3.4 Training Objective

3.4 训练目标

We perform contrastive learning by applying three distinct loss functions. Since our focus is on the quality of the manifold representation rather than emotion classification alone, we utilize the embeddings e directly during evaluation. Consequently, we employ neither additional linear classification layers nor cross-entropy loss. This approach ensures that the distinct effects of each loss function are directly reflected in the characteristics of the embedding space. We employ the Supervised InfoNCE REvisited (SINCERE) loss (Feeney and Hughes, 2023) as our baseline to achieve maximum discriminability while avoiding the problematic intra-class repulsion inherent in the original Supervised Contrastive Loss (Khosla et al., 2020). It computes the mean loss across all positive pairs within the batch B of size $B := |B|$ for a given anchor as follows:

我们通过应用三种不同的损失函数来进行对比学习。由于我们的关注点在于流形表示的质量，而不仅仅是情绪分类，因此我们在评估过程中直接使用嵌入向量 e 。因此，我们既不使用额外的线性分类层，也不使用交叉熵损失。这种方法确保了每种损失函数的独特效果能够直接反映在嵌入空间的特性中。我们采用监督信息最大化（SINCERE）损失（Feeney 和 Hughes, 2023）作为基准，以避免原始监督对比损失（Khosla 等人，2020）中固有的类内排斥问题的同时实现最大的判别性。该损失函数在给定锚点的情况下，计算批次 B （大小为 $B := |B|$ ）中所有正样本对的平均损失，具体如下：

$$\mathcal{L}_{\text{SINCERE}} = \frac{1}{B} \sum_{i \in \mathcal{B}} \left(\frac{-1}{|\mathcal{P}|} \sum_{j \in \mathcal{P}} \log \frac{\exp(e_i^T e_j / \tau)}{Z_i} \right) \quad (5)$$

where τ is a temperature, $\mathcal{P}$ is the in-batch positive set of the anchor sentence x_i , and Z_i represents the term corresponding to the positive sample as well as the in-batch negative samples:

其中 τ 是温度， $\mathcal{P}$ 是锚句 x_i 的同一批次正样本集合，而 Z_i 表示与正样本以及同一批次负样本对应的项：

$$Z_i = \exp(e_i^T e_j / \tau) + \sum_{k \in \mathcal{N}} \exp(e_i^T e_k / \tau) \quad (6)$$

where $\mathcal{N}$ is the in-batch negative set. Since this loss function computes the loss for all pairs within the batch, it induces a strong separation between positive and negative samples. However, because it applies equal weight to all negative samples, it

其中 $\mathcal{N}$ 是小批量内的负样本集合。由于这个损失函数计算了小批量内所有样本对的损失，它在正样本和负样本之间产生了强分离。然而，由于它对所有负样本赋予相同的权重，因此

fails to account for the specific degree of separation required between the anchor and each individual negative sample. Given that the ECM positions similar emotions nearby and opposing emotions at antipodes, the loss function should ideally dictate specific pairwise distances. Accordingly, we utilize SoftCSE for soft constraints and CircularCSE for hard constraints. We refine the ‘weight individualization’ method proposed in previous SoftCSE research, as it offers a more intuitive formulation and greater numerical stability (Zhuang et al., 2024). SoftCSE assigns individual weights to the negative sample terms in the SINCERE loss:

未能考虑到锚点与每个单独的负样本之间所需的特定分离程度。鉴于 ECM 将相似的情绪放置在附近，将对立的情绪放置在相对的两端，损失函数理想情况下应规定特定的成对距离。因此，我们使用 SoftCSE 来实现软约束，并使用 CircularCSE 来实现硬约束。我们改进了之前 SoftCSE 研究中提出的“权重个性化”

方法，因为它提供了更直观的表述和更高的数值稳定性（Zhuang 等，2024）。SoftCSE 在 SINCERE 损失中为负样本项分配了个体权重：

$$Z_i = \exp(e_i^T e_j / \tau) + \sum_{k \in \mathcal{N}} w_{ik} \exp(e_i^T e_k / \tau), \quad (7)$$

$$w_{ik} = \frac{1 - \cos(\Delta\theta_{ik})}{\frac{1}{|\mathcal{N}|} \sum_{k \in \mathcal{N}} (1 - \cos(\Delta\theta_{ik}))} \quad (8)$$

where $\Delta\theta_{ik}$ is the angular difference on the ECM. The numerator of w_{ik} is inversely proportional to the similarity of the emotion labels i and k i.e., $w_{ik} \propto -\cos(\Delta\theta_{ik})$, meaning that the closer the labels are on the circle, the smaller the weight and the weaker the repelling force becomes. The denominator acts as a normalization factor within the batch, adjusting the scale of the negative terms to match Equation (6) i.e., $\sum_{k \in \mathcal{N}} 1 = \sum_{k \in \mathcal{N}} w_{ik} = |\mathcal{N}|$. While the original paper employed $e_i^T e_k$ computed via a frozen encoder model instead of $\cos(\Delta\theta_{ik})$, this approach incurs additional inference overhead during training and relies heavily on the performance of the encoder. Therefore, we pre-define pairwise distances based on the ECM. This design facilitates the straightforward assignment of individual weights to negative samples.

其中 $\Delta\theta_{ik}$ 是在 ECM 上的角度差异。 w_{ik} 的分子与情绪标签 i 和 k 的相似性成反比，即 $w_{ik} \propto -\cos(\Delta\theta_{ik})$ ，这意味着标签在圆上越接近，权重越小，排斥力越弱。分母则在批次内充当归一化因子，调整负项的尺度以匹配方程 (6) 即 $\sum_{k \in \mathcal{N}} 1 = \sum_{k \in \mathcal{N}} w_{ik} = |\mathcal{N}|$ 。虽然原始论文使用通过冻结编码器模型计算的 $e_i^T e_k$ 而不是 $\cos(\Delta\theta_{ik})$ ，但这种方法在训练过程中会带来额外的推理开销，并且高度依赖编码器的性能。因此，我们基于 ECM 预定义成对距离。这种设计便于对负样本进行直接的权重分配。

As an even stronger constraint, we propose CircularCSE, which directly learns distances on the circle:

作为一种更严格的约束，我们提出了 CircularCSE，它直接在圆上学习距离：

$$\mathcal{L}_{\text{CircularCSE}} = \frac{1}{B(B-1)} \sum_{i,j \in \mathcal{B}: i \neq j} \ell_{ij},$$

$$\ell_{ij} = \begin{cases} [\max(0, |e_i^T e_j - \cos(\Delta\theta_{ij})| - m)]^2 & \text{if } y_i = y_j \\ (e_i^T e_j - \cos(\Delta\theta_{ij}))^2 & \text{otherwise} \end{cases} \quad (9)$$

Here, $m > 0$ is a margin hyperparameter that allows for tolerance within classes. Although a margin is desirable to account for variations in nuance and intensity among samples sharing the same label, it inevitably lowers the model’s discriminability.

在这里， $m > 0$ 是一个边界超参数，它允许在类别内部有一定的容忍度。虽然一个边界对于考虑到具有相同标签的样本之间在细微差别和强度上的变化是有益的，但它不可避免地会降低模型的可区分性。

4 Experiments

4 实验

4.1 Experimental Setup

4.1 实验设置

Datasets. Our experiments utilize three real-world datasets: Emolit (Rei and Mladenici, 2023), Empathetic Dialogue (Rashkin et al., 2019), SuperEmotion (de Fortuny, 2025), and one synthetic dataset: PersonaGen (Inoshita and Harada, 2025). Featuring a wide range of emotion labels, these datasets allow us to evaluate the geometric fidelity of the embedding space by analyzing how emotional categories are structurally organized. We select samples annotated with labels that match or closely resemble those in the ECM shown in Figure 3. For each dataset, the training set consists of 500 instances sampled per emotion label (450 for SuperEmotion), and the test set comprises 100 instances per label. Detailed descriptions of each dataset and the construction pipeline for the synthetic dataset are provided in Appendix C.

数据集. 我们的实验使用了三个真实世界数据集: Emolit (Rei 和 Mladenici, 2023), Empathetic Dialogue (Rashkin 等, 2019), SuperEmotion (de Fortuny, 2025), 以及一个合成数据集: PersonaGen (Inoshita 和 Harada, 2025)。这些数据集具有广泛的情绪标签, 使我们能够通过分析情绪类别在结构上的组织方式, 评估嵌入空间的几何保真度。我们选择与图 3 中所示 ECM 中标签匹配或非常接近的标注样本。对于每个数据集, 训练集包含每种情绪标签下采样的 500 个实例 (SuperEmotion 为 450 个), 测试集包含每种标签下 100 个实例。每个数据集的详细描述以及合成数据集的构建流程见附录 C。

Models. To account for variations in embedding space properties resulting from diverse architectures and pre-training objectives, we categorize the models into three groups, selecting two representatives from each: BERT-like Encoders (mE5 (Wang et al., 2024) and mxbai (Lee et al., 2024)), LLM-based Encoders (Qwen3-Embedding-4B (Zhang et al., 2025) and Llama-Embed-Nemotron-8B (Babakhin et al., 2025)), and Decoder-only LLMs (Llama-3.2-3B (Grattafiori et al., 2024) and OLMo-3-7B (OLMo Team et al., 2025)).

模型. 为了考虑由于不同的架构和预训练目标导致的嵌入空间属性的变化, 我们将模型分为三组, 每组选择两个代表性模型: BERT-like 编码器 (mE5 (Wang 等, 2024) 和 mxbai (Lee 等, 2024))、基于大语言模型 (LLM) 的编码器 (Qwen3-Embedding-4B (Zhang 等, 2025) 和 Llama-Embed-Nemotron-8B (Babakhin 等, 2025)), 以及仅解码器的大语言模型 (Llama-3.2-3B (Grattafiori 等, 2024) 和 OLMo-3-7B (OLMo 团队, 2025))。

Implementation Details. We attach a single-layer Transformer Block (GPT head) or a normalized Transformer Block (nGPT head) to the final layer of each backbone model. The head shares the same attention mask, dimension size, and attention head count as the backbone. Final sentence embeddings are derived via specific pooling methods: the [CLS] token for BERT-like encoders, the last token for decoder-only LLMs and Qwen3-Embedding. For Llama-Embed-Nemotron, we utilize mean pooling, consistent with its original training methodology. During training, BERT-like encoders are fully fine-tuned, while the LLM backbones remain frozen with only the heads being trained. Further implementation details are provided in Appendix D.

实现细节. 我们将在每个主干模型的最终层上附加一个单层 Transformer Block (GPT 头) 或一个归一化 Transformer Block (nGPT 头)。该头与主干模型共享相同的注意力掩码、维度大小和注意力头数量。最终的句子嵌入是通过特定的池化方法得到的: 对于 BERT 类编码器, 使用 [CLS] token; 对于仅解码器的 LLM 和 Qwen3-Embedding, 使用最后一个 token。对于 Llama-Embed-Nemotron, 我们采用平均池化, 与它的原始训练方法一致。在训练过程中, BERT 类编码器会完全微调, 而 LLM 主干模型保持冻结, 仅对头进行训练。更多实现细节请参见附录 D。

4.2 Evaluation Metrics

4.2 评估指标

We assess how well the trained embeddings capture emotional expressions via clustering analysis. Following the protocol of the MTEB clus-

我们通过聚类分析评估训练后的嵌入向量在多大程度上捕捉了情感表达。按照 MTEB 聚类分析的协议，

Training Objective	Head Arch.	mE5		Qwen3-Embedding-4B		Llama-3.2-3B	
V_{Measure}	CD-r	V_{Measure}	CD-r	V_{Measure}	CD-r		
Pretrained		0.342	0.574	0.495	0.522	0.094	0.217
SINCERE	- GPT	0.760	0.317	0.756	0.305	0.725	0.358
- nGPT	0.744	0.221	0.739	0.545	0.577	0.425	
SoftCSE	- GPT	0.755	0.477	0.751	0.552	0.710	0.548
- nGPT	0.753	0.499	0.723	0.708	0.516	0.728	
CircularCSE	- GPT	0.717	0.757	0.643	0.747	0.579	0.728
- nGPT	0.720	0.764	0.659	0.753	0.382	0.708	

培训目标	首席建筑师	mE5		Qwen3-Embedding-4B		Llama-3.2-3B	
V_{Measure}	CD-R	V_{Measure}	CD-R	V_{Measure}	CD-R		
预训练		0.342	0.574	0.495	0.522	0.094	0.217
真诚	- GPT	0.760	0.317	0.756	0.305	0.725	0.358
- nGPT	0.744	0.221	0.739	0.545	0.577	0.425	
SoftCSE	- GPT	0.755	0.477	0.751	0.552	0.710	0.548
- nGPT	0.753	0.499	0.723	0.708	0.516	0.728	
圆周率 CSE	- GPT	0.717	0.757	0.643	0.747	0.579	0.728
- nGPT	0.720	0.764	0.659	0.753	0.382	0.708	

Table 1: Summary of average performance across all datasets for selected models. Best results per model (excluding Pretrained) are bolded, and worst are underlined. V_{Measure} indicates clustering quality, and CD-r indicates correlation with the circumplex distance.

表 1: 所选模型在所有数据集上的平均性能总结。每个模型的最佳结果（不包括预训练模型）以粗体表示，最差结果以下划线表示。 V_{Measure} 表示聚类质量，CD-r 表示与环形距离的相关性。

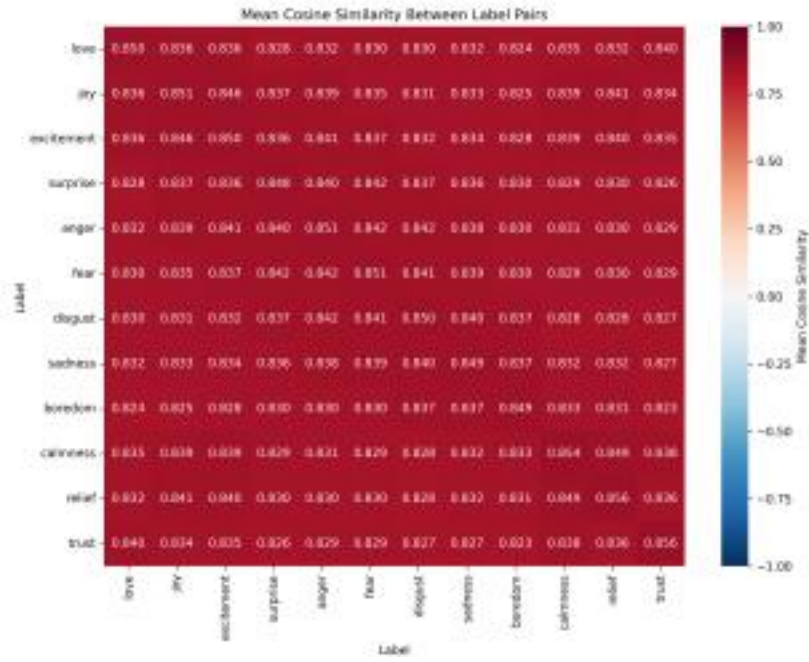


Figure 3: images/964163598ed81ce4c4a8e772e10640e6a58af5199b972a99c02f34b664d07917.jpg

(a) Pretrained

(a) 预训练

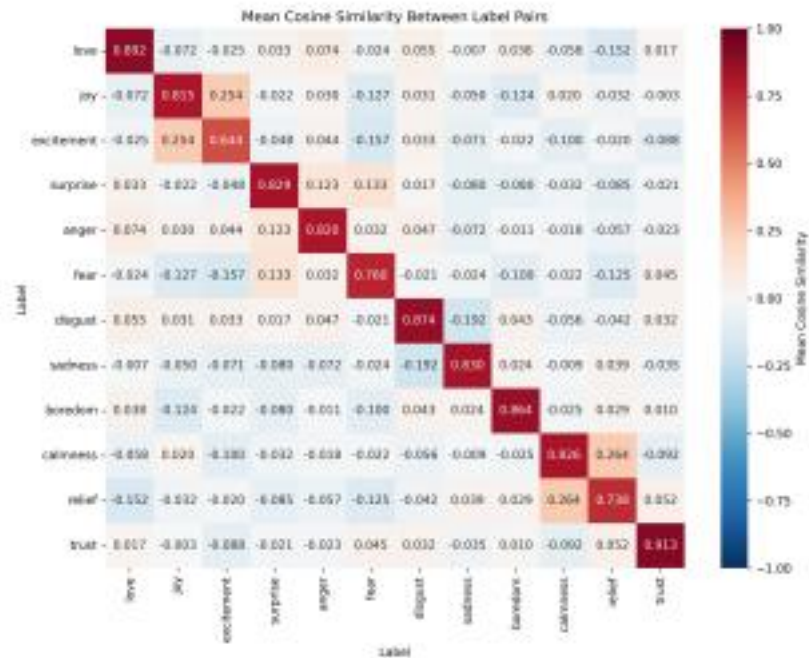


Figure 4: images/78c15f0ded5bf8a072496063e2af5821addb86302075e9db6be4df3d192b9c36.jpg

(b) SINCERE-nGPT

(b) SINCERE-nGPT

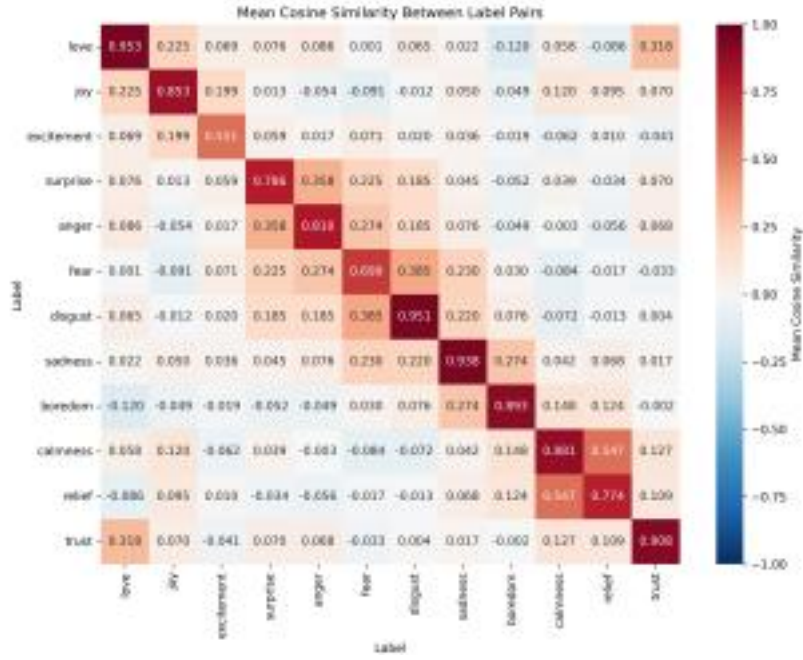


Figure 5: images/a08abfbd1bf45b8dd6e9f6e292e61db01b16a344ea72b40d467d497fd53a8fa4.jpg

(c) SoftCSE-nGPT

(c) SoftCSE-nGPT

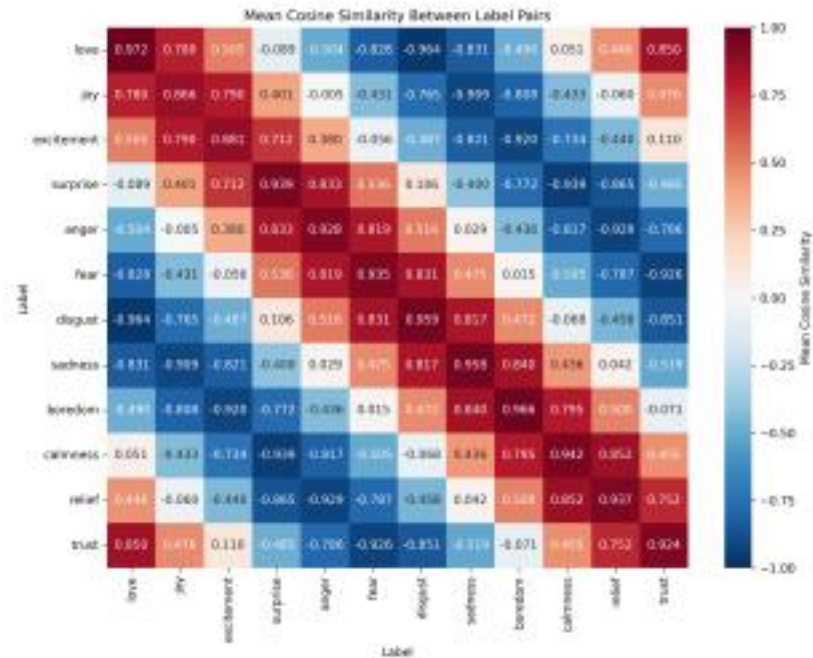


Figure 6: images/fcf84f40d198a5bee05f9088995820095029f4610017cfc1ed89c045a2787f97.jpg

(d) CircularCSE-nGPT

Figure 4: Average cosine similarity between emotion label pairs of mE5

(d) CircularCSE-nGPT

图 4: mE5 情绪标签对的平均余弦相似度

tering tasks (Muennighoff et al., 2023), we partition the test set into clusters using k-means (MacQueen, 1967), setting the number of clusters k equal to the number of ground truth labels, and evaluate performance using V-Measure (Rosenberg and Hirschberg, 2007). Given that our models are optimized for cosine similarity, we utilize Spherical k-means (Dhillon and Modha, 2001), which uses cosine distance instead of the Euclidean ℓ_2 metric. The algorithm is run 10 times with varied initializations, and the result with the minimum inertia is selected. We evaluate not only the discriminative power of the embedding space but also the extent to which its structure aligns with human perception. Building upon the approaches of (Zhao et al., 2016, 2024a), we define the Circumplex Distance (CD) on the ECM as follows: $CD(y_i, y_j) := C + \text{AngleDistance}(y_i, y_j)$ where C represents the constant inter-polarity distance, and $\text{AngleDistance}(y_i, y_j)$ denotes the number of steps between labels on the ECM. We define the distance between Neutral and Positive/Negative polarities as 2, the distance between Positive and Negative polarities as 4, and the distance between identical polarities as 0. This configuration ensures that the distance between opposing polarities con-

在进行聚类任务 (Muennighoff 等人, 2023), 我们使用 k-means (MacQueen, 1967) 将测试集划分为簇, 设置簇的数量 k 等于真实标签的数量, 并使用 V-Measure (Rosenberg 和 Hirschberg, 2007) 评估性能。鉴于我们的模型是为余弦相似度优化的, 我们使用球面 k-means (Dhillon 和 Modha, 2001), 该方法使用余弦距离而不是欧几里得 ℓ_2 度量。算法使用不同的初始化方式运行 10 次, 选择惯性最小的结果。我们不仅评估嵌入空间的判别能力, 还评估其结构与人类感知的匹配程度。在 (Zhao 等人, 2016, 2024a) 的方法基础上, 我们在 ECM 上定义了环形距离 (CD) 如下: $CD(y_i, y_j) := C + \text{AngleDistance}(y_i, y_j)$ 其中 C 表示恒定的极性间距离, $\text{AngleDistance}(y_i, y_j)$ 表示 ECM 上标签之间的步数。我们将中性与积极/消极极性之间的距离定义为 2, 积极与消极极性之间的距离定义为 4, 相同极性之间的距离定义为 0。这种配置确保了对立极性之间的距离为 2。

sistently exceeds the distance between identical polarities. By incorporating the premise that polarity differences are cognitively more significant than mere steps on the circle, CD reflects human psychological emotion structure more faithfully. To measure the alignment with this metric, we propose the Pearson correlation with CD (CD-r):

始终超过相同极性之间的距离。通过结合极性差异在认知上比圆周上的简单步骤更为重要的前提, CD 更真实地反映了人类心理情感结构。为了衡量与这一指标的一致性, 我们提出了与 CD 的皮尔逊相关系数 (CD-r):

$$CD - r := \text{Pearson}(CD(y_i, y_j), 1 - \text{AvgCosSim}(y_i, y_j)) \quad (10)$$

$$\text{AvgCosSim}(y_i, y_j) := \frac{1}{N_i N_j} \sum_{k=1}^{N_i} \sum_{l=1}^{N_j} e_k^T e_l \quad (11)$$

with N_i and N_j representing the total number of samples for each emotion label in the test set.

其中 N_i 和 N_j 表示测试集中文档中每种情绪标签的样本总数。

4.3 Results

4.3 结果

Table 1 presents the average V-Measure and CD-r across datasets for mE5, Qwen3-Embedding-4B, and Llama-3.2-3B (comprehensive results are provided in Table 4 of Appendix E).

表 1 展示了 mE5、Qwen3-Embedding-4B 和 Llama-3.2-3B 在不同数据集上的平均 V-Measure 和 CD-r (完整结果请参见附录 E 的表 4)。

Regarding the training objectives, SINCERE and SoftCSE yield higher V-Measure scores, whereas CircularCSE underperforms. Conversely, for CD-r, SINCERE scores are low, while CircularCSE achieves higher results. This phenomenon can be intuitively explained by referencing Figure 4,

关于训练目标，SINCERE 和 SoftCSE 产生的 V-Measure 分数更高，而 CircularCSE 表现较差。相反，在 CD-r 指标上，SINCERE 的得分较低，而 CircularCSE 则取得了更高的结果。这种现象可以通过参考图 4 来直观地解释，

which illustrates the average cosine similarity between emotion label pairs for each head. Pre-trained models typically form an anisotropic embedding space, resulting in high similarity across all emotion label pairs (Ethayarajh, 2019). In contrast, SINCERE attempts to position negative samples orthogonally, driving similarities toward zero. SoftCSE relaxes this constraint, and CircularCSE aligns the similarity of each emotion label pair with the corresponding cosine similarity on the circular manifold. Since clustering requires high discriminative power between labels, SINCERE is advantageous; however, it fails to sufficiently capture the relational structure between labels, leading to a lower CD-r. While CircularCSE successfully captures the ordinal relationships between labels, it makes distinguishing between adjacent labels more difficult, resulting in a lower V-Measure. These results highlight a fundamental conflict between the objectives of deep learning, which prioritizes discriminative accuracy, and psychology, which emphasizes alignment with human perception, creating a trade-off between accuracy and interpretability.

这展示了每个头的平均余弦相似度在情感标签对之间的关系。预训练模型通常形成各向异性的嵌入空间，导致所有情感标签对之间的相似度较高 (Ethayarajh, 2019)。相比之下，SINCERE 试图将负样本正交地放置，从而将相似度推向零。SoftCSE 放松了这一约束，而 CircularCSE 则将每个情感标签对的相似度与圆环流形上的相应余弦相似度对齐。由于聚类需要标签之间的高判别性，SINCERE 具有优势；然而，它未能充分捕捉标签之间的关系结构，导致 CD-r 较低。而 CircularCSE 成功捕捉了标签之间的顺序关系，但它使区分相邻标签变得更加困难，导致 V-Measure 较低。这些结果突显了深度学习的目标（优先考虑判别准确性）与心理学（强调与人类感知的一致性）之间存在根本性的冲突，从而在准确性与可解释性之间形成权衡。

Revisiting Table 1, we observe distinct trends across models; notably, for Llama-3.2-3B, the nGPT model exhibits a lower V-Measure. This suggests that Decoder-only models rely more heavily on vector norms to encode contextual information compared to Encoder models, implying that the normalization process leads to information loss. Furthermore, the significantly low accuracy of the pretrained models indicates that while emotion information is latent within the embeddings, it is not explicitly separable; instead, the embedding space prioritizes contextual information or features required for next-token prediction.

回顾表 1，我们观察到不同模型之间呈现出明显的趋势；值得注意的是，对于 Llama-3.2-3B 模型，nGPT 模型表现出较低的 V-Measure 值。这表明，与编码器模型相比，仅解码器的模型更依赖于向量范数来编码上下文信息，这意味着归一化过程会导致信息丢失。此外，预训练模型的准确率显著偏低，这表明虽然情感信息潜藏在嵌入中，但并未明确可分离；相反，嵌入空间优先考虑的是用于下一个词预测的上下文信息或特征。

5 Analysis

5 分析

5.1 Conflict between Psychological and Deep Learning Models

5.1 心理学模型与深度学习模型之间的冲突

To better elucidate the differences between the individual methods, we conduct further analysis. Figures 1(b), (c), and (d) visualize the results of applying PCA to the Emolit test set embeddings for each mE5 head. For SINCERE-nGPT and SoftCSE-nGPT, although the emotions separate into distinct clusters, the structures lack clear regularity, and the explained variance ratio of the principal components is low. In contrast, CircularCSE-nGPT exhibits a clear circular arrangement of emotions, with a significantly higher explained variance ratio.

为了更好地阐明不同方法之间的差异，我们进行了进一步的分析。图 1(b)、(c) 和 (d) 展示了对 Emolit 测试集嵌入向量应用 PCA 后，每个 mE5 头的结果可视化。对于 SINCERE-nGPT 和 SoftCSE-nGPT，虽然情绪能够分离成不同的簇，但结构缺乏明显的规律性，主成分的解释方差比率较低。相比之下，CircularCSE-nGPT 表现出情绪的明显环形排列，其解释方差比率显著更高。

Hypothesizing that this difference in arrangement would manifest clearly under distinct scenarios, we conducted the following two experiments:

假设这种排列方式差异在不同场景下会表现得非常明显，我们进行了以下两个实验：

1. Robustness to dimensionality reduction.
2. Robustness to the number of emotion labels.
3. 对降维的鲁棒性。
4. 对情绪标签数量的鲁棒性。

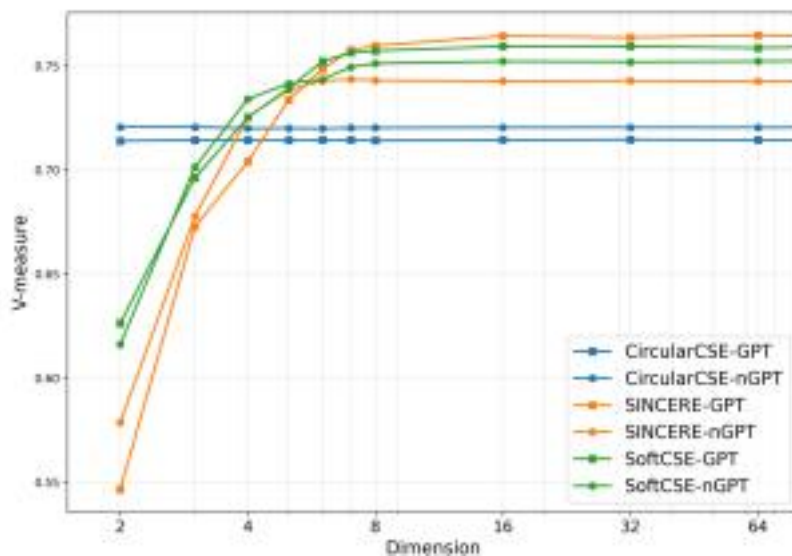


Figure 7: images/5ea722bd19afc5efd337c0ccb2a24725df6b8cb0ac789c1cb074b8693cd8351c.jpg

(a) Robustness to dimensionality reduction

(a) 对降维的鲁棒性

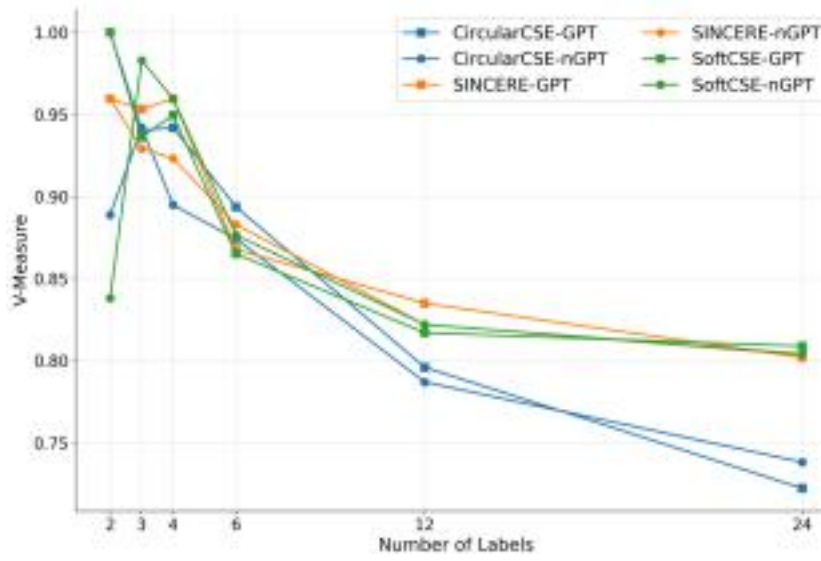


Figure 8: images/7eeeb2365bdf895e424c2e6bb3710dd2df44bd4d1f505f52b932ab1c4a47f292.jpg

(b) Robustness to number of labels

Figure 6: Clustering performance of mE5 heads under different conditions. (a) Impact of PCA dimensionality reduction. (b) Impact of the number of emotion labels.

(b) 对标签数量的鲁棒性

图 6: 在不同条件下, mE5 头的聚类性能。(a) PCA 降维的影响。(b) 情绪标签数量的影响。

The experimental results are presented in Figure 6. Figure 6(a) shows the average V-Measure across datasets when clustering with Spherical k-means after reducing the dimensions of each mE5 head via PCA. Figure 6(b) illustrates the change in V-Measure on the Emolit dataset as the number of emotion labels varies (results for other models and details on label configurations are provided in Appendix F). CircularCSE maintains stable accuracy even in low dimensions and performs comparably to SINCERE and SoftCSE when the number of emotion labels is small. However, its performance degrades in high-dimensional settings or with a large number of labels. These results can be explained by the optimal solutions and optimal margins of the respective loss functions. The SINCERE loss function, given by Equations (5) and (6), attains its theoretical lower bound when the cosine similarity between all positive-negative pairs equals $\frac{-1}{E-1}$ (Proof in Appendix G). In practice, however, due to the curse of dimensionality, the positive-negative similarity often settles at a local optimum of 0, resulting in an average boundary margin of 90° (orthogonality). In contrast, since CircularCSE arranges labels on a ring, the geometry of its optimal solution is consistently a 2-dimensional circle, and the boundary margin between any positive-negative pair is at most $\frac{\pi}{E}$ (for a 12-class classification problem, the maximum margin is 30°). The optimal boundary margins of SINCERE and CircularCSE coincide only in 2 dimensions; otherwise, the gap in discriminative power widens as the model dimensionality or the number of labels increases. This indicates that arranging emotions on a circular manifold, i.e., attempting to make representations visualization-friendly or se-

实验结果如图 6 所示。图 6(a) 展示了在使用球形 k 均值聚类时, 通过 PCA 降低每个 mE5 头的维度后, 各数据集的平均 V-Measure。图 6(b) 说明了在 Emolit 数据集上, 随着情绪标签数量的变化, V-Measure 的变化情况 (其他模型的结果和标签配置的详细信息请参见附录 F)。CircularCSE 在低维情况下仍能保持稳定的准确性, 并且在情绪标签数量较少时, 其表现与 SINCERE 和 SoftCSE 相当。然而, 当在高维设置或标签数量较多时, 其性能会下降。这些结果可以通过相应损失函数的最优解和最优边距来解释。根据公式 (5) 和 (6) 定义的 SINCERE 损失函数, 在所有正负对之间的余弦相似度等于 $\frac{-1}{E-1}$ 时, 达到其理论下限

(证明请参见附录 G)。然而，在实践中，由于维度灾难的影响，正负相似度通常会收敛于 0 的局部最优值，从而导致平均边界边距为 90° （正交性）。相比之下，由于 CircularCSE 将标签布置在环上，其最优解的几何形状始终是二维圆，任意正负对之间的边界边距最大为 $\frac{\pi}{E}$ （对于 12 类分类问题，最大边距为 30° ）。SINCERE 和 CircularCSE 的最优边界边距仅在二维情况下一致；否则，随着模型维度或标签数量的增加，其判别能力的差距会变大。这表明，将情绪布置在圆环流形上，即尝试使表示可视化友好或 se-

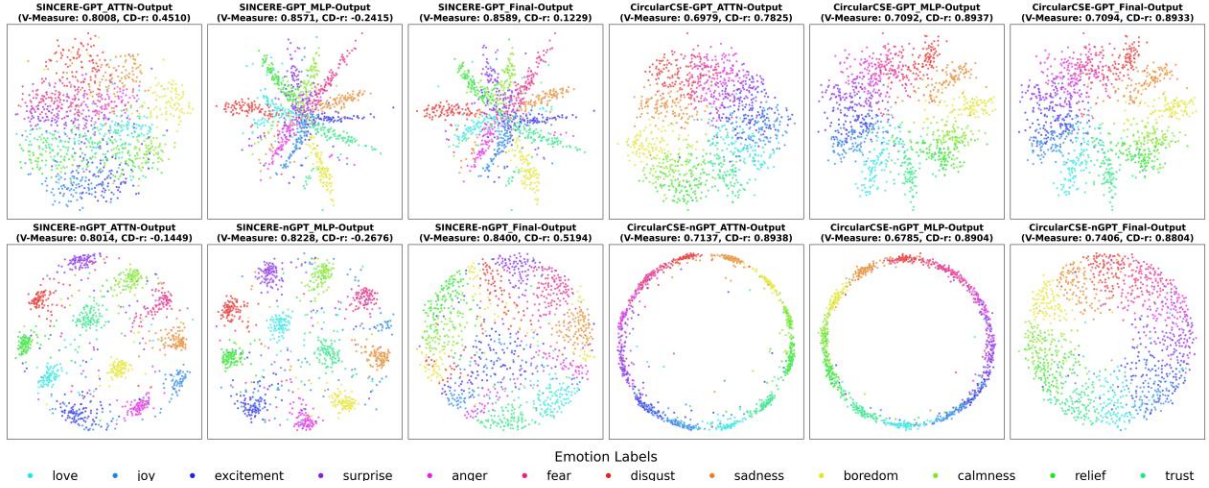


Figure 9: images/78265ce005d7adc765ddaa2d3246c813f9a4d3447bf0d5affa43da4d4c3b7420.jpg

Figure 5: Visualization of embedding representations from each module of the GPT and nGPT heads using Multidimensional Scaling (MDS). Additional results for other models are presented in Appendix H.

图 5：使用多维尺度（MDS）对 GPT 和 nGPT 头部每个模块的嵌入表示进行可视化。其他模型的额外结果见附录 H。

manically interpretable, implicitly imposes a low-dimensional manifold structure, which conflicts with the high-dimensional representations typical of deep learning. This limitation becomes more pronounced as model capacity increases (higher dimensionality) or as task demands rise (distinguishing among a diverse set of emotions).

语义可解释性隐式地施加了一个低维流形结构，这与深度学习中典型的高维表示相冲突。这种限制在模型容量增加（更高的维度）或任务需求增加（区分多样情感）时变得更加明显。

5.2 Effects of Spherical Constraints

5.2 球形约束的影响

We perform a qualitative evaluation of the head architectural differences using dimensionality reduction. Specifically, we extract intermediate representations from within the transformer block of Qwen3-Embedding-4B and employ Multidimensional Scaling (MDS) for dimensionality reduction (de Leeuw, 2005), followed by clustering and visualization. Since MDS arranges points to reconstruct the pairwise distance matrix of the samples, it is particularly effective at preserving global geometry in high-dimensional spaces.

我们使用降维技术对头部架构差异进行定性评估。具体来说，我们从 Qwen3-Embedding-4B 的 transformer 块中提取中间表示，并采用多维标度（MDS）进行降维（de Leeuw, 2005），随后进行聚类 and 可视化。由于 MDS 通过排列点来重构样本的成对距离矩阵，因此在高维空间中特别擅长保留全局几何结构。

The results are presented in Figure 5. The impact of the spherical constraint is prominently re-

flected in the structural differences of the representations. In the GPT head, the output of the MLP layer dominates the final result; the MLP amplifies differences in magnitude (norm), effectively overwriting prior intermediate representations with vectors of larger norms. Additionally, clusters exhibit linear elongation, where concepts appear as directional basis vectors. Although this expansion increases intra-cluster variance—allowing for a wider range of captured representations—it hinders Euclidean-based classification, as points may inadvertently lie closer to external clusters than to their own.

结果如图 5 所示。球形约束的影响在表示结构的差异中得到了明显体现。在 GPT 头部，MLP 层的输出主导了最终结果；MLP 放大了幅度（范数）的差异，有效地用范数更大的向量覆盖了先前的中间表示。此外，聚类表现出线性拉伸，其中概念以方向基向量的形式出现。尽管这种扩展增加了簇内方差，从而允许捕捉更广泛的表示，但它妨碍了基于欧几里得的分类，因为点可能无意中更接近外部簇而非自身的簇。

In contrast, in the nGPT head, the removal of

相比之下，在 nGPT 头部中，移除

the norm component results in the Attention and MLP layers exhibiting similar output geometries, with clusters adopting complex, non-linear structures rather than straight lines. We attribute this to the fact that on the hypersphere, differences between concepts are expressed solely via angular components rather than norms, causing directions to encode composite rather than singular concepts.

范数组件导致注意力层和 MLP 层表现出相似的输出几何结构，聚类采用复杂的非线性结构，而不是直线。我们归因于这一现象的原因在于，在超球面上，概念之间的差异仅通过角度分量来表达，而不是范数，导致方向编码的是复合概念而非单一概念。

These findings suggest that even when quantitative evaluation metrics appear identical, the underlying representation structures can differ vastly. Manifold-aware research remains scarce, and determining the optimal manifold structure for specific tasks is an open question for future work.

这些发现表明，即使定量评估指标看起来相同，其底层表示结构也可能存在显著差异。与流形相关的研究仍然较为匮乏，确定特定任务的最佳流形结构仍然是未来工作的开放性问题。

6 Discussion and Conclusion

6 讨论与结论

We compared conventional models with those reproducing psychological circular structures by performing contrastive learning in Euclidean and spherical spaces. Through this comparison, we discovered an unavoidable structural dilemma: a trade-off exists between discriminative performance and human interpretability. This highlights the importance of selecting the appropriate method based on the intended application. Although this study focused on clustering, our framework is transferable to representation learning for other concepts or tasks. For instance, interpretable approaches can potentially be applied to tasks such as identifying underrepresented labels and verifying the validity of emotion placement (Figure 14 in Appendix H provides an example). Moreover, as interpretability and visualization facilitate model controllability, extending this approach to generation tasks is a key direction for future research.

我们通过在欧几里得空间和球面空间中进行对比学习，将传统模型与能够再现心理循环结构的模型进行了比较。通过这种比较，我们发现了一个不可避免的结构困境：判别性能与人类可解释性之间存在权衡。这突显了根据预期应用场景选择适当方法的重要性。尽管本研究重点在于聚类，但我们的框架可以迁移到其他概念或任务的表示学习中。例如，可解释的方法可以潜在地应用于诸如识别未被充分表示的标签和验证情绪位置有效性等任务（附录 H 中的图 14 提供了一个示例）。此外，由于可解释性和可视化有助于模型的可控性，将这种方法扩展到生成任务是未来研究的关键方向。

Limitations

限制

Task Simplification. We simplified the task formulation for our experiments. Real-world emotions are highly complex and often involve simultaneous conflicting states (e.g., bittersweet, calm anger). However, in this study, we operate under the assumption that each text corresponds to a single emotion label. Furthermore, regarding the “Neutral” state, while we treated it as a distinct category belonging to neither Positive nor Negative extremes, it is intrinsically located at the center (origin) of Russell’s circumplex model. Integrating these complex cases into our current framework would require specialized mechanisms; developing a more natural representation for such states remains a challenge for future work.

任务简化。我们简化了实验中的任务定义。现实中的情绪高度复杂，通常涉及同时存在的矛盾状态（例如，苦乐参半、平静的愤怒）。然而，在本研究中，我们假设每段文本对应一个单一的情绪标签。此外，关于“中性”状态，尽管我们将其视为一个独立的类别，不属于积极或消极的极端，但它本质上位于 Russell 的环形模型的中心（原点）。将这些复杂情况整合到我们当前的框架中，需要专门的机制；为这些状态开发更自然的表示方式仍然是未来工作的挑战。

Modality Constraints. Emotion analysis spans not only text but also modalities such as images and audio. Certain emotional states (e.g., boredom or drowsiness) are more readily manifested through gestures or non-verbal cues rather than text. Capturing these nuances necessitates a multimodal architectural design.

模态约束。情感分析不仅涉及文本，还包括图像和音频等模态。某些情感状态（例如无聊或困倦）更倾向于通过手势或非语言线索来表现，而不是文本。捕捉这些细微差别需要多模态的架构设计。

Connection to Psychological Models. Although the circumplex model used in this study references Russell’s model, some emotion placements differ from the original configuration to align with the specific labels available in our dataset. Additionally, numerous circumplex models exist beyond Russell’s proposal, and it is possible that our specific arrangement does not perfectly reflect the true relationships between emotions (discussed in Appendix C.4). However, since our conclusions are largely driven by the dimensionality of the embedding space, we anticipate obtaining similar results with other 2- or 3-dimensional visualizable psychological models. Reproducing higher-dimensional and more complex emotional models is an objective for future research.

与心理模型的联系。尽管本研究中使用的环形模型参考了 Russell 的模型，但某些情绪的位置与原始配置有所不同，以适应我们数据集中可用的具体标签。此外，除了 Russell 的提议之外，还存在许多其他的环形模型，因此我们的特定安排可能并未完全反映情绪之间的真正关系（在附录 C.4 中讨论）。然而，由于我们的结论在很大程度上由嵌入空间的维度驱动，我们预计使用其他二维或三维可视化的心理模型也能获得类似的结果。重现更高维度和更复杂的情绪模型是未来研究的目标。

Ethical Considerations

伦理考量

This study primarily utilizes publicly available and properly licensed datasets and models, all of which permit research use. The sole exception is the generation of synthetic data, where we employed both open-source and proprietary models to create a new dataset. While this dataset may potentially contain harmful content, it was used exclusively for training classification tasks; consequently, the trained models themselves pose no safety risks. All datasets and models were used in accordance with

本研究主要使用公开可用且已获得适当授权的数据集和模型，所有这些数据集和模型均允许用于研究。唯一的例外是合成数据的生成，我们在其中使用了开源和专有的模型来创建一个新的数据集。虽然该数据集

可能包含有害内容，但仅用于分类任务的训练；因此，训练后的模型本身不会带来安全风险。所有数据集和模型的使用均遵循相关规定。

their intended research purposes.

其预期的研究目的。

We used AI tools for proofreading and mathematical typesetting in the preparation of this paper. All content has been thoroughly reviewed and verified by the human authors.

我们使用了 AI 工具进行论文的校对和数学公式排版。本文的所有内容都已由人类作者进行了彻底的审查和验证。

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